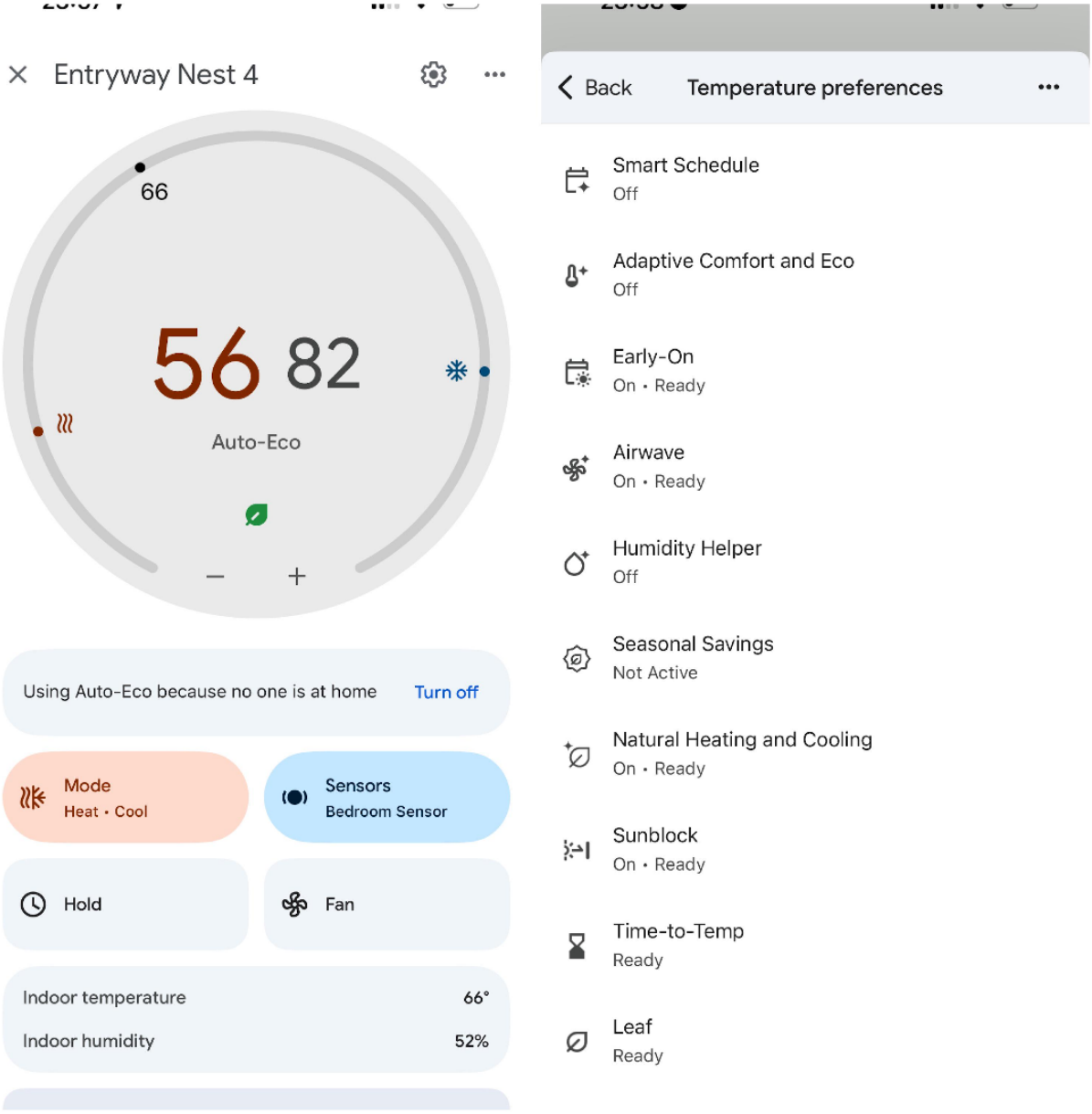


Figure 1.11 The interface to a Nest smart thermostat showing how it takes action based on your physical location. Eco mode is enabled based on presence sensing via the location of mobile phones in the household. Image © Thomas Smailus, Ph.D. P.E. Reproduced with permission.

At this point, your thermostat is more than just a remote control for your AC system. It's running a simplified digital twin of your home's thermal behavior. It knows how long it takes to warm up in the morning, how sunlight through your west-facing windows affects the living room, and when nobody's home. It's constantly comparing what should be happening (according to its model) with what is happening (from real-time sensors), and it adjusts accordingly, all without your input.